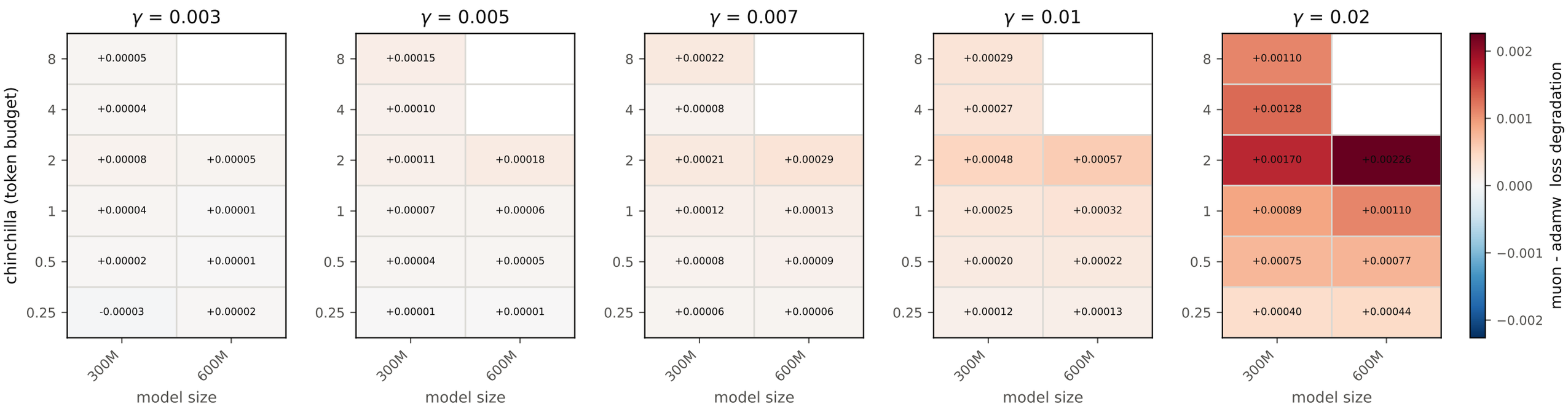
Degradation under Gaussian weight perturbation: muon vs adamw ($\gamma \leq 0.02$), 300M and 600M
γ = 0.003
γ = 0.005
γ = 0.007
γ = 0.01
γ = 0.02
chinchilla (token budget)
8
4
2
1
0.5
0.25
+0.00005
+0.00004
+0.00008
+0.00005
+0.00004
+0.00001
+0.00002
+0.00001
-0.00003
+0.00002
+0.00015
+0.00010
+0.00011
+0.00018
+0.00007
+0.00006
+0.00004
+0.00005
+0.00001
+0.00001
+0.00022
+0.00008
+0.00021
+0.00029
+0.00012
+0.00013
+0.00008
+0.00009
+0.00006
+0.00006
+0.00029
+0.00027
+0.00048
+0.00057
+0.00025
+0.00032
+0.00020
+0.00022
+0.00012
+0.00013
+0.00110
+0.00128
+0.00170
+0.00226
+0.00089
+0.00110
+0.00075
+0.00077
+0.00040
+0.00044
300M
600M
model size
muon - adamw loss degradation
0.002
0.001
0.000
−0.001
−0.002
cell = (perturbed - unperturbed) for muon minus the same for adamw | negative (blue) = muon degrades less